SMOPS 2019 Round 1

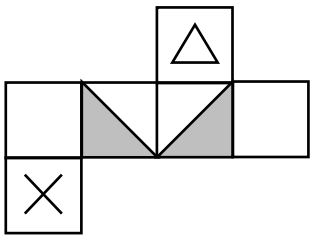
Time: 120 minutes

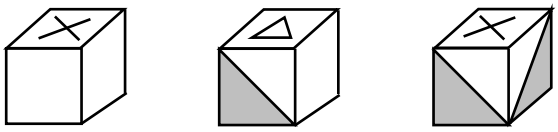
1. In the addition equation below, A and B represent two different numbers. Find the value of $A + B$.

$$\begin{array}{r} AB \\ BA \\ +\quad B \\ \hline AAB \end{array}$$

2. $a^2 + b^2 \leq 10$, where a and b are both positive integers and not necessarily distinct. How many possible pairs of (a, b) are there? For example, (1, 2) and (2, 1) are considered different pairs.

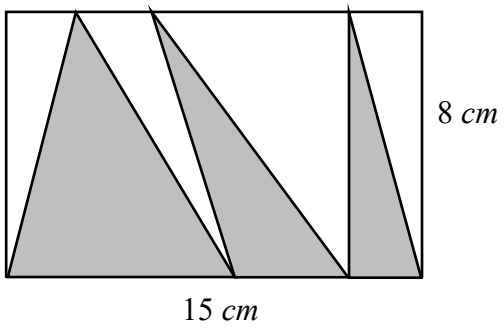
3. The diagram below can be folded into one of the three figures (1), (2), (3). Find the correct cube.





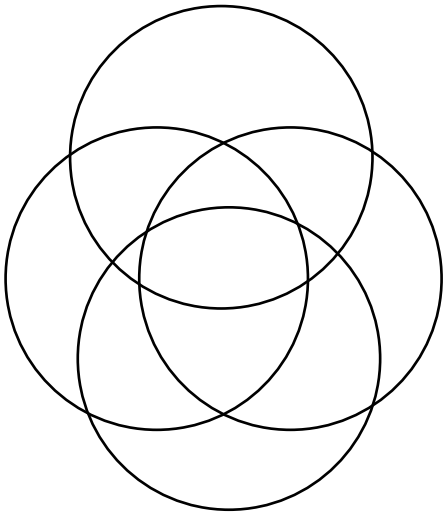
(1) (2) (3)

4. As shown in the diagram, the length and breadth of the rectangle are 15 cm and 8 cm respectively. Find the total area of the shaded regions.



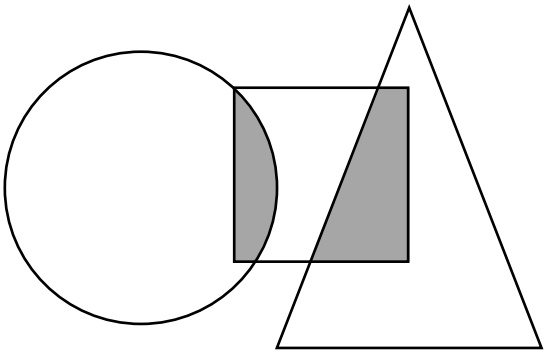
5. Calculate: $(2019 + 101) \times (2019 + 99) - (200 + 2019) \times 2019$.

6. In the diagram on the right, fill in the 13 portions that are split by the four circles with these 13 numbers 1, 2, 3, …, 12, 13. Then, add the 7 numbers in each circle. Lastly, add the four sums of each circle to obtain a final total sum. What is the biggest possible sum?

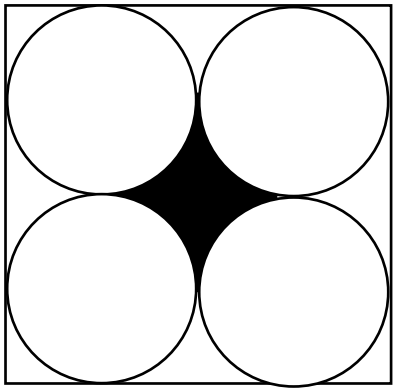


7. Given that $A = 0.123456789101112 \ldots 998999$, where the number behind the decimal point is formed by writing the numbers 1 to 999 in succession. What number is at the 2019th position on the right of the decimal point?

8. In the diagram on the right, the shaded area in the circle is $\frac{1}{6}$ of the total area of the circle and is $\frac{1}{5}$ of the area of the square. The shaded area in the triangle is $\frac{1}{9}$ of the area of the triangle and is $\frac{1}{4}$ of the area of the square. The ratio of the area of the circle to the square and to the triangle is $24: a: b$, where a and b are whole numbers. Find the value of $a + b$.



9. As shown in the diagram, four identical circles are surrounded by a square of area $1600\ \text{cm}^2$. Find the area of the shaded region. (Take $\pi = 3.14$)



10. A whole number N is exactly divisible by 2, 3, 4, 5, 6, 8, 9, and the remainder is 5 when N is divided by 7. Find the smallest possible value of N.

11. Arrange the odd numbers 1, 3, 5, 7, … into 5 columns according to the table below. If the number "2019" appears in the mth row and the nth column from the left, find the value of $m+n$.

	1	3	5	7
15	13	11	9	
	17	19	21	23
31	29	27	25	
	…	…	…	…
…	…	…	…	

12. In a 9×9 square table, there are 29 small squares are dyed black. If x represents the number of rows that contain at least 5 small black squares and y represents the number of columns that contains at least 5 small black squares, find the largest possible value of $x+y$.

13. $\frac{79}{137}$ is represented in the form below

$$a-\cfrac{1}{b+\cfrac{1}{c-\cfrac{1}{d+\cfrac{1}{e}}}}$$

Given that a, b, c, d, e are whole numbers, find the value of $a+b+c+d+e$.

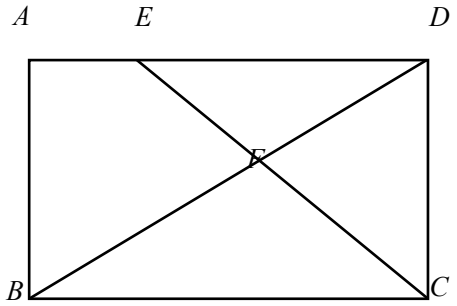
14. A bag has 2 cards, one card has 2 red sides, the other card has one red side and one blue side. One card is picked at random from the bag and placed on a table. If the side facing upwards is red, what is the probability that the side facing downwards is also red? Choose the correct answer below.

(1) 1 (2) $\frac{3}{4}$ (3) $\frac{2}{3}$ (4) $\frac{1}{2}$

15. Divide the numbers 1059, 1417 and 2312 by the positive integer d. The remainders obtained are all r, where r is a whole number greater than 1. Find the value of $d - r$.

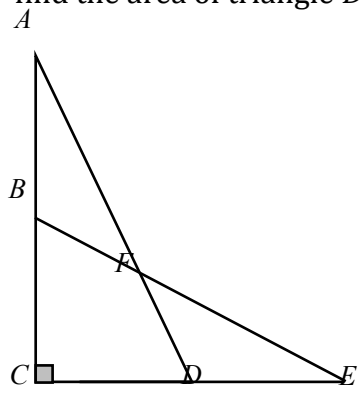
16. Find the sum of all the digits in these 999 numbers 1, 2, 3, ..., 998, 999.

17. As shown in the diagram, BD and CE split rectangle $ABCD$ into four parts. The area of triangle DEF is $16\ \text{cm}^2$, the area of triangle CDF is $20\ \text{cm}^2$。 Find the area of quadrilateral $ABFE$.

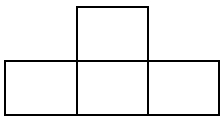


18. Choose two different numbers from the group of numbers {19, 20, 21, ..., 98, 99}, such that their sum is an even number. How many ways are there?

19. As shown in the diagram, AC is perpendicular to CE, D is the midpoint of CE, B is the midpoint of AC. If AD and EB intersect at F, and $BC = CD = 30$ cm, find the area of triangle DEF.



20. Take out one T-shape made up of 4 small squares from an 8×8 square chess board (the shape can be rotated). How many ways are there?



21. Several football teams take part in a football competition. Two teams play one match. For each match, the winner gets 3 points, loser gets one point. If the match ends in a draw, each team gets 2 points. After the competition, the total points of all the teams is 24. How many teams participated in the competition?

22. Given that $x = \frac{1}{2} \times \frac{3}{4} \times \frac{5}{6} \times \ldots \times \frac{47}{48}$, $y = \frac{2}{3} \times \frac{4}{5} \times \frac{6}{7} \times \ldots \times \frac{48}{49}$, $z = \frac{1}{7}$, if these three numbers are arranged from smallest to biggest, which answer below is correct?

(1) $z < y < x$ (2) $z < x < y$ (3) $x < z < y$
(4) $x < y < z$ (5) $y < z < x$

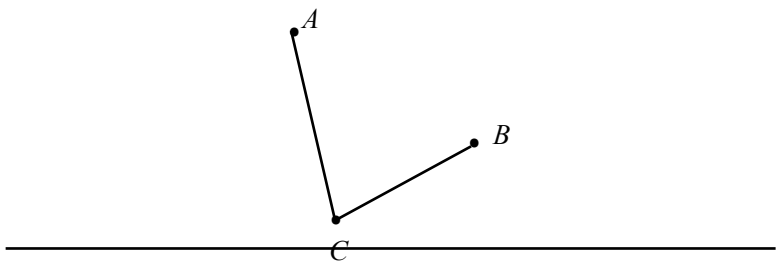
23. When a 3-digit number is divided by the sum of its digits, the quotient is q. What is the smallest possible value of q?

24. A bag contains three different coloured balls red, white and blue. It is known that the number of blue balls is at least half of the number of white balls and at most $\frac{1}{3}$ of the number of red balls. Also, the total number of white and blue balls is 55. What is the least number of red balls in the bag?

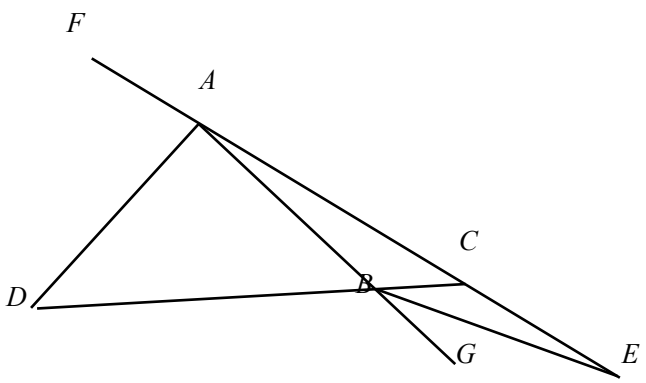
25. The perimeter of a circular track is 20 metres. Arthur and Thomas set off at the same time and walk around the track. Arthur walks non-stop at a uniform speed of 1.5m/s around the track in the clockwise direction. Thomas walks non-stop at a uniform speed of 3.5m/s around the track in the anti-clockwise direction. If both of them walk sufficiently long, how many different points along the circular track will they meet?

26. There are 10 questions in a math competition. Competition rules dictates as such: 4 points are awarded for every correct answer; 0 points are awarded for every unanswered question while 1 point will be deducted for a wrong answer. How many possible different results are there for this math competition?

27. As shown in the diagram below, there is a 8-metre long line segment. Points A and B are on the same side of the line segment. It is known that the line between points A and B is 2 metres, i.e. $AB = 2$ metres. The perpendicular distances from points A and B to the line segment are 2 metres and 0.5 metres respectively. Now, John wants to find a point C on the line segment, such that the difference between line segments AC and BC is the greatest. What is the largest possible difference of line segments AC and BC?



28. As shown in the diagram, F is point on CA extended, G a point on AB extended. The angle bisector of $\angle FAB$ meets CB extended at point D. The angle bisector of $\angle CBG$ meets AC extended at point E. If $AD = AB = BE$, find the value of $\angle BAC$.



29. Calculate: $\frac{(2^2+4^2+\cdots+98^2+100^2)-(1^2+3^2+\cdots+97^2+99^2)}{50}$.

30. Choose several numbers from these 50 numbers {1, 2, 3, ..., 49, 50} such that the sum of any two random numbers chosen is not exactly divisible by 7. How many numbers can be chosen at most?